

Rajarata University of Sri Lanka,
Department of ICT.

ICT 1209 – Web Technologies

AURAFIT

Fitness Tracker

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Project Overview

AuraFit is a responsive, web-based fitness tracking and wellness management platform developed using HTML5, CSS3, Bootstrap 5, Vanilla JavaScript, PHP 8, and MySQL. Designed to consolidate fragmented health tracking into a single ecosystem, the application provides an integrated user experience where individuals can securely manage profiles, log daily workouts, record water and calorie consumption, monitor body weight, and calculate their Body Mass Index (BMI). AuraFit also incorporates curated meal and nutrition planning, alongside dynamic dashboards featuring visual charts and analytics to track physical progress across desktop, tablet, and mobile interfaces.

Purpose of the Application

- **Centralize Health & Fitness Data:** Eliminates the inconvenience of switching between multiple standalone apps by offering a unified space for activity logs, dietary records, and metric tracking.
- **Encourage Consistency & Motivation:** Transforms raw fitness data into interactive charts and graphs, helping users visualize weight trends, workout history, and BMI improvements over time.
- **Facilitate Goal-Oriented Nutrition:** Guides users toward healthy dietary habits by pairing calorie monitoring with structured meal and nutrition plans tailored to their personal wellness targets.
- **Deliver Accessible Wellness Management:** Ensures busy professionals, beginners, and fitness enthusiasts can manage and update their health journey seamlessly across any modern device.

Target Audience

- **Fitness Beginners:** Individuals taking their first steps in personal health and exercise who require a straightforward, user-friendly tool to calculate their Body Mass Index (BMI) and establish baseline habits.
- **General Fitness Enthusiasts:** Active users looking for a consolidated platform to log regular workouts, monitor body weight variations, and systematically track their calorie and water intake.

- **Busy Professionals and Working Individuals:** People with demanding schedules who need an efficient, unified dashboard to log daily health data quickly without juggling multiple standalone tools.
- **Health-Conscious Individuals:** Anyone aiming to maintain a balanced lifestyle through structured meal plans, consistent nutritional tracking, and long-term wellness monitoring.

Key Features Implemented

1. Authentication System
 - **User Registration & Profile Creation:** Collects user credentials and initial health metrics (age, height, weight, fitness goals) through a validated form, storing records in the MySQL database.
 - **Session-Based Login:** Utilizes PHP sessions (`$_SESSION`) to authenticate users, protect private dashboard routes, and manage user state across visits.
 - **Security & Password Protection:** Incorporates server-side sanitization and PHP password hashing (`password_hash`) to ensure secure credential storage and prevent unauthorized access.
2. Contact Form
 - **Inquiry Interface:** A dedicated front-facing form allowing users to submit messages, general questions, or feedback.
 - **Direct Form Processing:** Client-side validation verifies field inputs before submission, handled via PHP post-processing to confirm submission status without permanent database archiving.
3. Theme-Specific Core Features & Tracking
 - **Fitness & Health Data Logging:** Direct form interfaces allowing users to log daily health metrics, including workout activity, body weight, calorie consumption, and daily water volume.
 - **Automated BMI Calculator:** An integrated evaluation module that calculates the user's Body Mass Index (BMI) using stored height and weight values to determine physical fitness status.
 - **Nutrition & Meal Plan Access:** Structured presentation of nutritional guidelines and curated meal plans tailored to support varied user fitness goals.
4. JavaScript Interactions & Data Visualization

- **Dynamic Chart.js Dashboard:** Leverages the Chart.js library to render visual analytics, plotting calorie history, weight tracking trends, and workout progress into line and doughnut charts.
- **Real-Time Client-Side Validation:** Instant input checks on form fields using Vanilla JavaScript to ensure numeric validity and complete fields before submitting to the backend.
- **Interactive UI Controls:** JavaScript-driven interface elements, including tab toggling between register/login views and responsive navigation controls across mobile and desktop breakpoints.

System Design

Database Schema

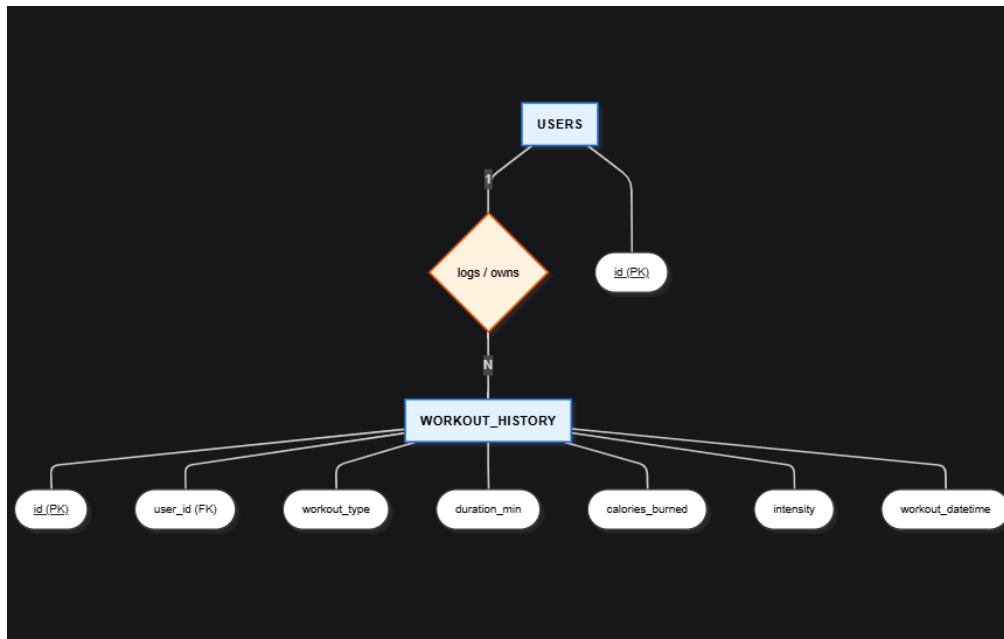


Figure 2: ER Diagram

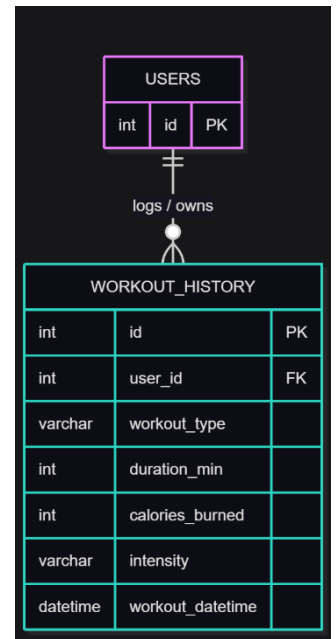


Figure 1: Database Schema

Folder Structure

1. aurafit/ (Root Project Directory)

1.1. css/

- 1.1.1. contact.css
- 1.1.2. content.css
- 1.1.3. dashboard.css
- 1.1.4. home.css
- 1.1.5. login.css

1.2. js/

- 1.2.1. about.js
- 1.2.2. dashboard.js
- 1.2.3. home.js
- 1.2.4. login.js

1.3. images/

1.4. html/

- 1.4.1. content.html
- 1.4.2. login.html

1.5. includes/

- 1.5.1. db.php
- 1.5.2. functions.php

1.6. auth/

- 1.6.1. register.php
- 1.6.2. login.php
- 1.6.3. logout.php

1.7. contact.php

1.8. index.php

1.9. dashboard.php

1.10. aurafit_db.sql

1.11. send_message.php

1.12. README.md

Implementation

Registration & Login

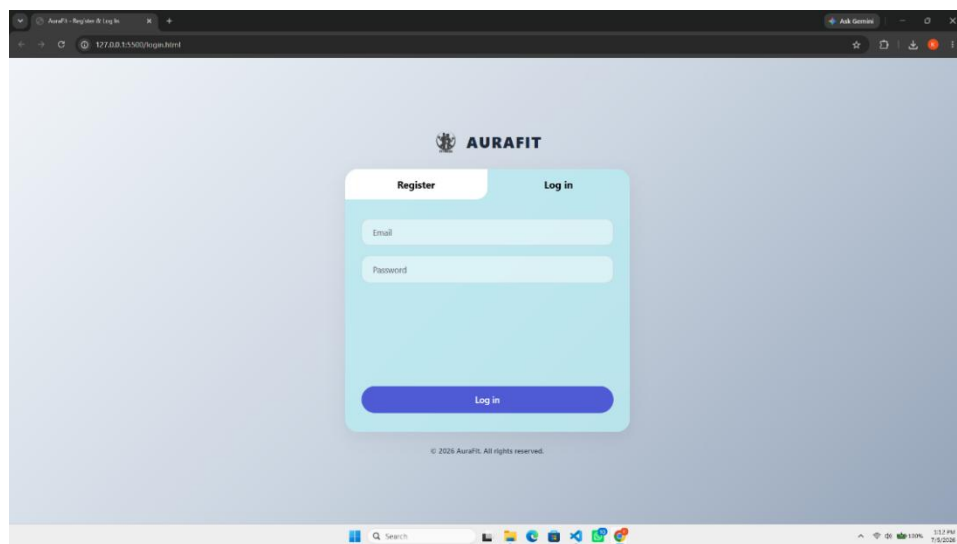


Figure 3: Login

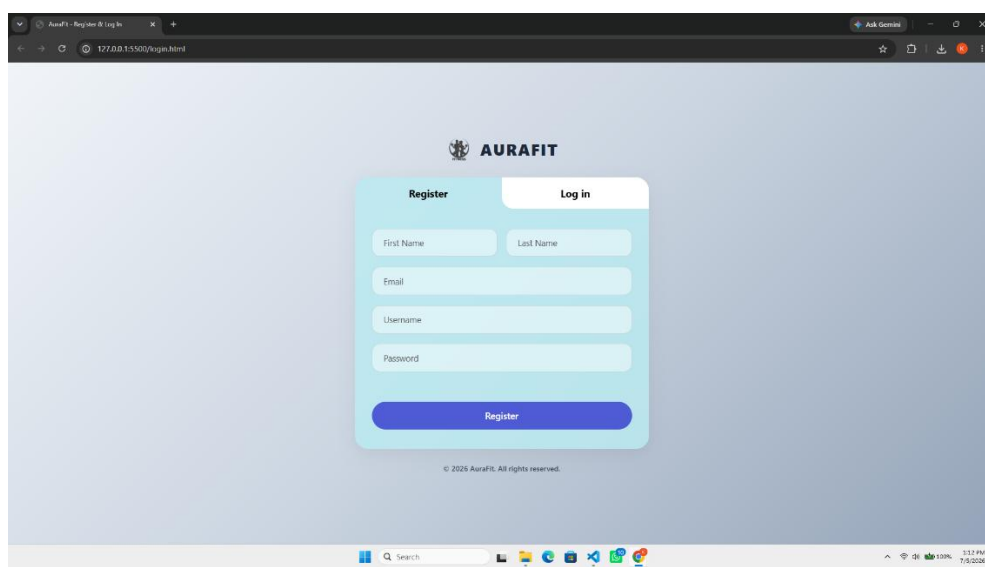
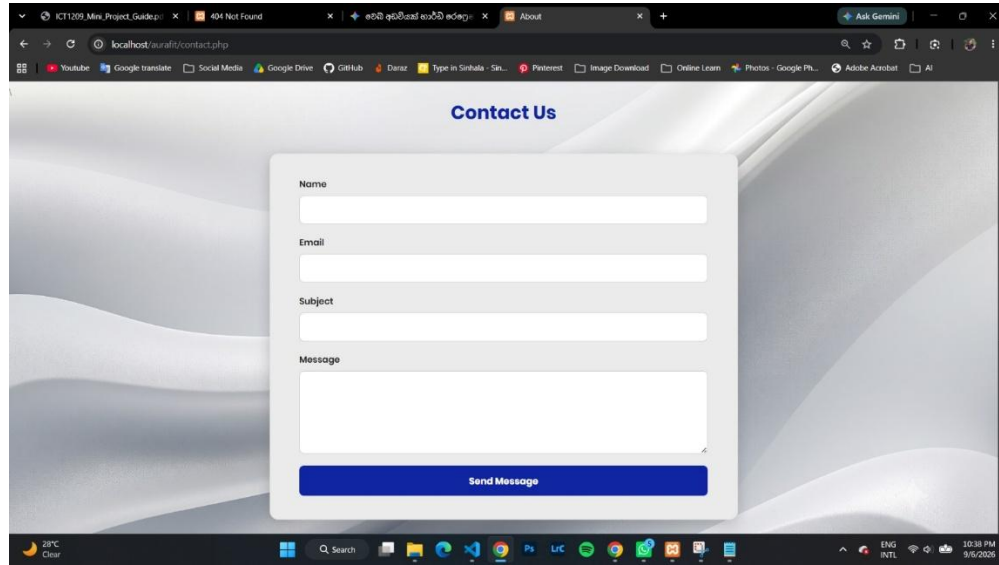


Figure 4: Register

Contact Form



The screenshot shows a web browser window with the URL `localhost/aurafit/contact.php`. The page has a light blue header with the text "Contact Us". Below the header is a white form with the following fields: "Name", "Email", "Subject", and "Message". A blue "Send Message" button is at the bottom of the form. The browser's address bar shows the URL, and the taskbar at the bottom displays various application icons and the system clock showing 10:38 PM on 9/6/2024.

Figure 5: Contact Form

Core Theme Features

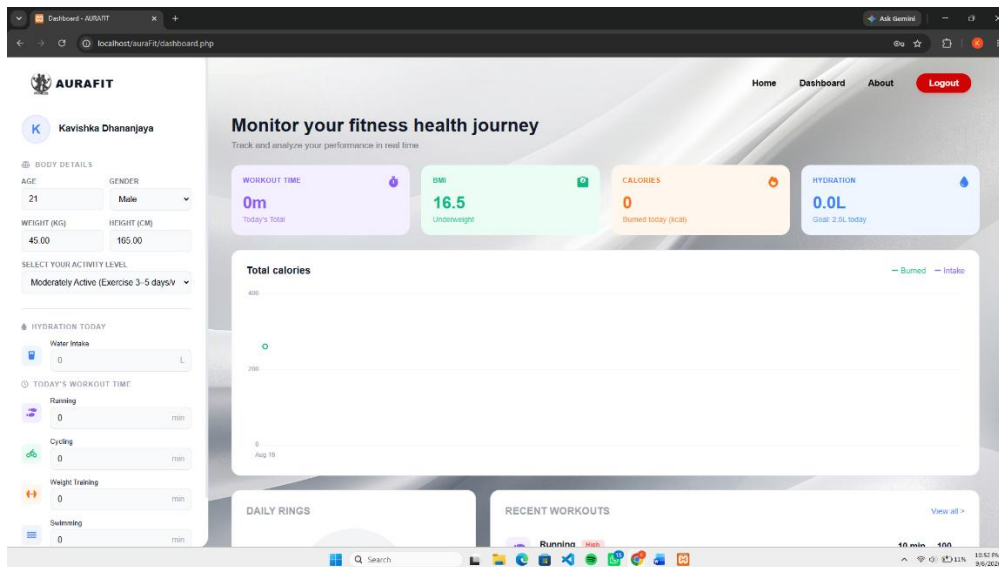


Figure 6: Total calories, Workout Time, BMI, Calories, Hydration

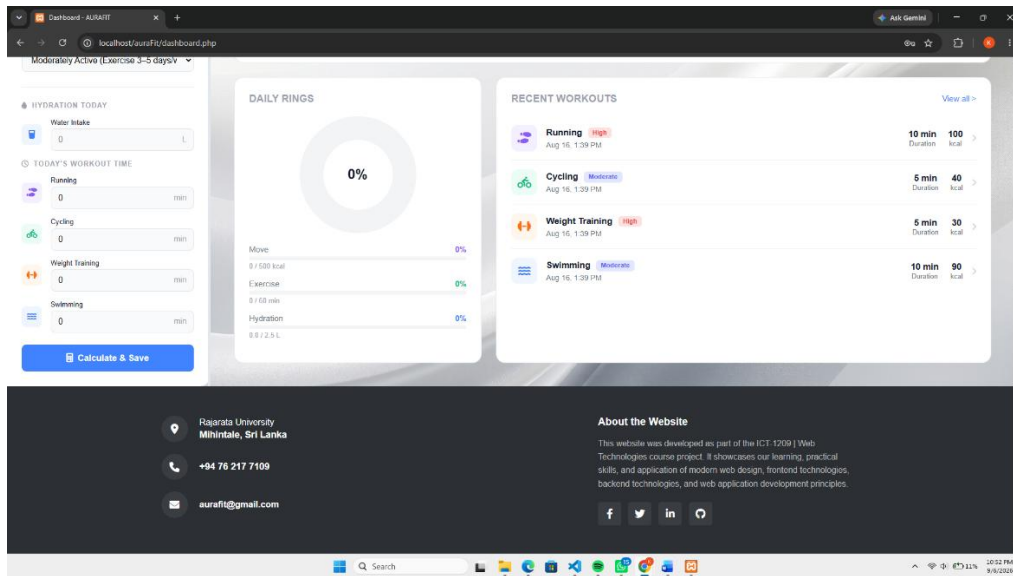


Figure 7: Daily Rings, Recent Workouts

Active JavaScript features

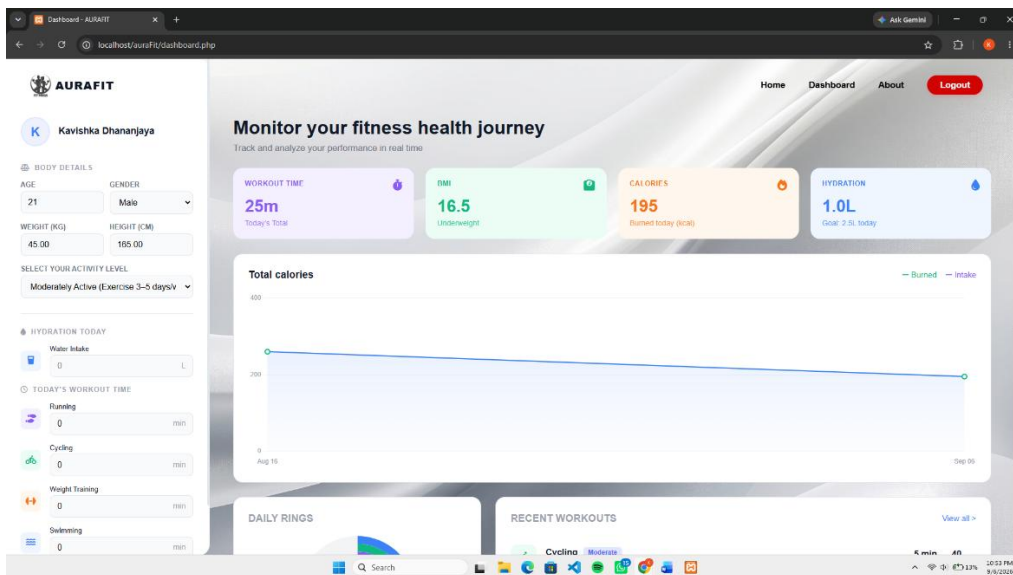


Figure 8: Active Javascript Features

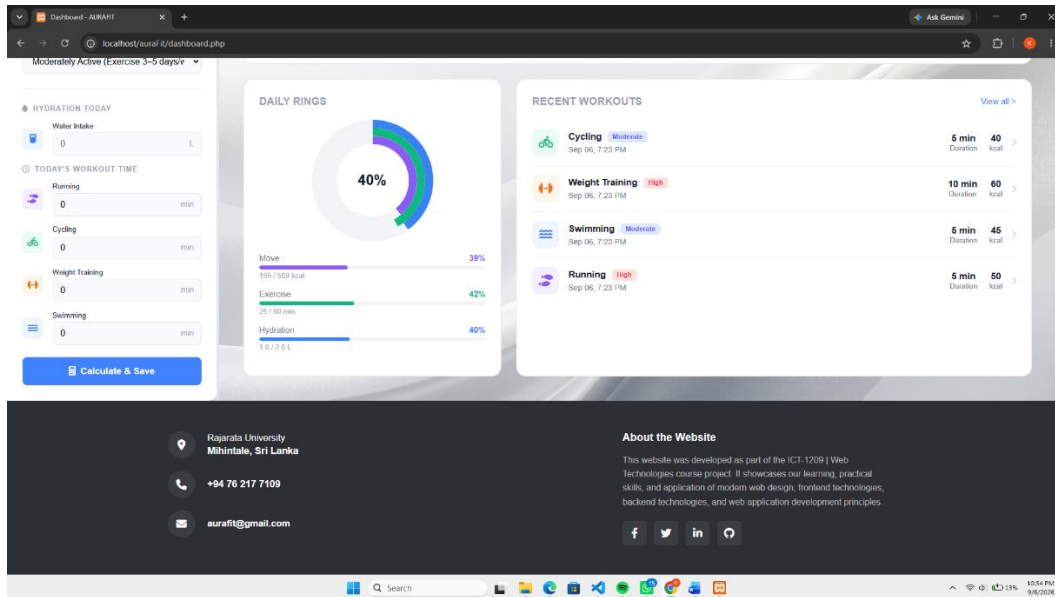


Figure 9: Active Javascript Features

Challenges and Solutions

1. Chart.js Canvas Resizing and Data Re-rendering Glitches

- Problem:** When rendering multiple Chart.js instances (line and doughnut charts) inside responsive Bootstrap layout grids, the charts failed to resize properly on viewport changes and occasionally rendered blank or duplicated canvases when updating logged metrics.
- Resolution:** Configured Chart.js with `responsive: true` and `maintainAspectRatio: false` within fixed-height parent containers. Implemented a utility function to check for existing chart instances using `Chart.getChart()` and `destroy (chart.destroy())` them before re-instantiating with new dataset arrays.

2. Session Fixation and Unauthorized Route Access

- Problem:** During user authentication testing, navigating directly to internal dashboard URLs allowed access or preserved old session tokens, presenting potential session fixation vulnerabilities and state inconsistencies across different browser tabs.
- Resolution:** Added `session_regenerate_id(true)` immediately upon successful credential validation in PHP to issue a fresh session ID and

invalidate the previous one. Additionally, created a centralized authentication check script (auth_guard.php) included at the top of all protected views to inspect \$_SESSION['user_id'] and redirect unauthenticated visitors to the login view.

3. SQL Injection Risks and Prepared Statement Binding

- **Problem:** Constructing dynamic SQL queries with raw string concatenation for multi-field forms (e.g., user registration handling credentials and health metrics like height, weight, and age) exposed the database to injection vulnerabilities and syntax errors caused by unescaped input.
- **Resolution:** Transitioned all database interactions from standard MySQL queries to PHP Data Objects (PDO) with parameterized prepared statements (\$stmt->prepare() and \$stmt->execute()). This ensured user inputs were treated strictly as data parameters rather than executable SQL logic, alongside applying server-side sanitization.

Individual Contribution

Table 1: Individual Contribution table

Member Name	Student Reg. No.	Contributions
D.S.Y.A.K.D.Abeythunga	ITT/2024/004	• Project repository initialization & file structure setup • Login & logout interface design, styling, and JS linking • Initial dashboard UI development (v1) • Database schema creation, configuration, and maintenance

M.T.P.N.Abeywardhana	ITT/2024/005	• Responsive Home, About, and Content page development • UI components (dynamic image slider, footer, dashboard cards) • Dashboard database connectivity & user session display • Frontend bug fixes & login error popups
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Setup Instructions

Deploy Project Files:

File Placement.

Clone or extract the AuraFit project folder into your local server's web root directory:

- For **XAMPP**: C:/xampp/htdocs/AuraFit

Start Server Modules: XAMPP Control Panel.

- Open the XAMPP Control Panel and start the **Apache** and **MySQL** services.

3.Create and Import Database: phpMyAdmin.

1. Open your web browser and go to <http://localhost/phpmyadmin/>.
2. Click **New** in the left sidebar to create a database (e.g.: aurafit_db).
3. Select the newly created database, navigate to the **Import** tab, and click **Choose File**.

Browse to the project folder, select aurafit_db.sql, and click **Import / Go**.

4. **Configure Database Connection: db.php**
Open the database configuration file (typically db.php) in a text editor and match your local MySQL credentials:

```
$host = 'localhost';  
  
$dbname = 'aurafit_db'; // Ensure this matches the database created in Step 3  
  
$username = 'root'; // Default XAMPP username  
  
$password = ""; // Default is blank (leave empty for XAMPP is often empty or  
'root')
```

5. **Launch the Application:** Web Browser.

Open a web browser and navigate to:

<http://localhost/AuraFit/>

References

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